

WASP-50b

R-1821

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_210905 · created 2026-08-03T04:45:20+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459462.914 +/- 0.055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.05
Transit depth (Rp/Rs)^2	3.03 +/- 1.75 [%]
Orbital Inclination (inc)	87.45 +/- 2.87
Airmass coefficient 1 (a1)	0.056 +/- 0.057
Airmass coefficient 2 (a2)	1.55 +/- 0.92
Scatter in the residuals of the lightcurve fit is	60.18 %
Best Comparison Star	#2 - [139, 200]
Optimal Method	PSF photometry
Transit Duration (day)	0.059 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.174 ± 0.05 vs 0.1404 (deviation 0.4σ)
Duration fit vs published	0.059 d vs 0.0754 d (deviation 0.34σ)
Mid-time O-C	-27.6 min
Depth SNR	2.1
Residual scatter	60.18 %

Light curve

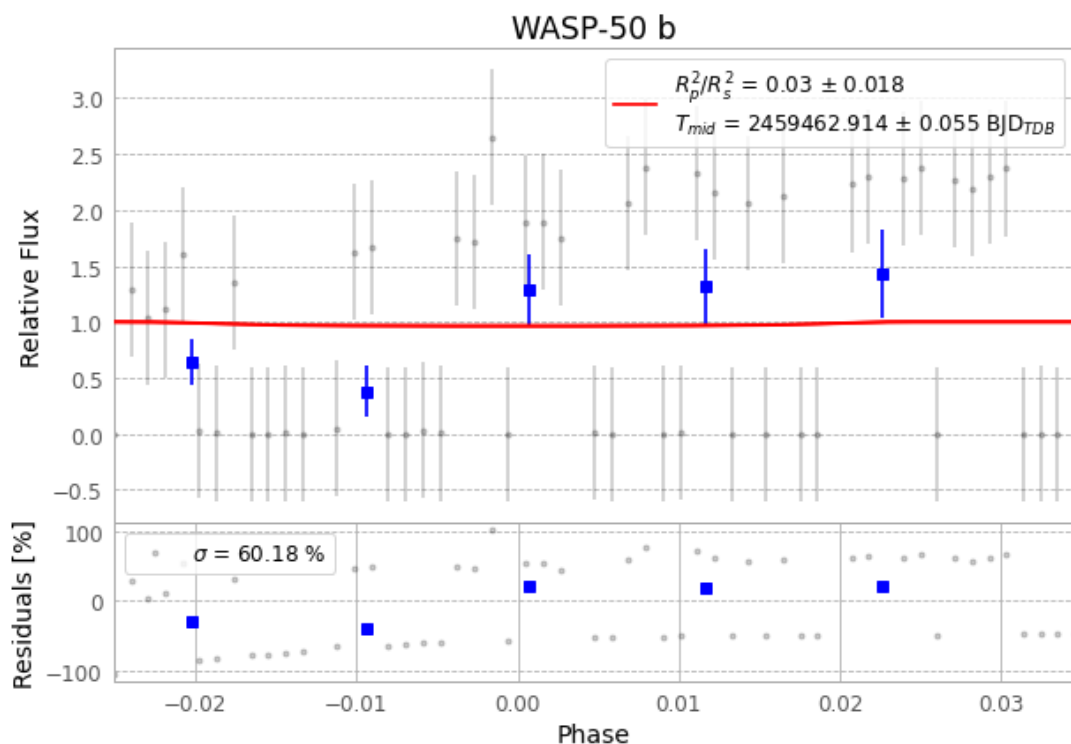


Figure 1 — FinalLightCurve_WASP-50 b_2021-09-05.png (SHA-256 d99e185af6e6c3f1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [358, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0d2646da11ff3361...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

54 FITS frames (35 MB), WASP-50210905083912.FITS → WASP-50210905112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-210905.log (196eee111ae2...), FinalLightCurve_WASP-50 b_2021-09-05.png (d99e185af6e6...), FinalLightCurve_WASP-50 b_2021-09-05.csv (6fadff1e54b8...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 04:45Z.